Web Technologies minimum Course Topics

Cover All the Lecture slides and cover these topics.

HTML

The HTML section begins with an OBE discussion and introduces the history of the Web, the Internet (WWW and URL), networking concepts including NAT and TCP/IP, web protocols, client-server architecture, the three-tier architecture, the HTTP protocol, differences between HTTP and HTTPS, web servers, types of servers, HTTP proxies, DNS, types of web browsers, and communication protocols. Students are introduced to HTML through basic syntax, HTML web page creation, forms, elements, attributes, page titles, favicons, block and inline elements, the HTML head section, tables, images, anchor tags, XML, and the Document Object Model (DOM). Laboratory activities include creating HTML web pages, developing registration forms using tables and HTML forms, understanding URL encoding and HTTP methods, and using Git for version control through repository creation, cloning, branching, push, and pull operations.

CSS

The CSS section introduces the fundamentals of styling web pages using CSS. Students learn about the five types of CSS selectors, layout and positioning techniques, the CSS box model, display properties, maximum width, overflow handling, z-index, and inline-block elements. Practical sessions focus on applying these concepts by designing and styling web forms while implementing layout techniques using relative, absolute, fixed, and sticky positioning to create structured, responsive, and visually appealing web pages

JavaScript

The JavaScript section introduces the history of JavaScript, different ways to apply JavaScript (inline, internal, and external), syntax, variables and operators, arrays, functions, objects, type casting, conditional statements, output methods, and string operations. Advanced topics include ES5 features, HTML DOM operations, style manipulation, event handling using action/event listeners, dynamically inserting, deleting, and updating HTML elements, and implementing client-side form validation. Laboratory exercises reinforce these concepts through JavaScript basics, DOM manipulation, event listeners, dynamic operations, and complete HTML form validation.